SCHOOL OF ELECTRONICS AND COMMUNICATION ENGINEERING

A MINI REPORT ON

PREDICTIVE MAINTENANCE OF INDUSTRIAL MOTORS USING MULTI SENSOR DATA

Submitted in fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY

IN
ELECTRONICS AND COMMUNICATION ENGINEERING

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DECLARATION

We, Rachana Shettar (R23EN111), Darshan Telsang (R23EN068), Mythri RN (R23EN091), students of B.Tech, School of Electronics and Communication Engineering, REVA University, declare that this Mini Project Report entitled "**Predictive Maintenance of Industrial Motors using Multi Data Sensors**" is the result of mini project work done by us under the supervision of Prof. Neethu KN Assistant Professor, School of ECE REVA University.

We are submitting this Report in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering by the REVA University, Bengaluru during the academic year 2025-26.

I declare that this project report satisfies the academic requirements concerning the mini-project work prescribed for the said Degree. I further declare that this mini project report or any part of it has not been submitted for the award of any other Degree/Diploma of this University or any other University/ Institution.

Signature of the Students:

Date: 15/05/2026

Certified that this mini project work submitted by Rachana Shettar(R23EN111), Darshan Telsang (R23EN040), Mythri RN (R23EN091), has been carried out under my guidance and the declaration made by the candidate is true to the best of my knowledge.

Signature of Guide:

Date: 15/05/2026

CERTIFICATE

Certified that the mini project work entitled "**Predictive Maintenance of Industrial Motors using Multi Data Sensors**" carried out under my guidance by **Rachana Shettar (R23EN111), Darshan Telsang (R23EN040), Mythri RN (R23EN091)** a bonafide student of REVA University during the academic year 2025-26, is submitting the mini project report in partial fulfillment for the award of **Bachelor of Technology** in **Electronics and Communication Engineering** during the academic year **2025-26.** The project report has been approved as it satisfies the academic requirements in respect of the Project work prescribed for the said Degree.

Signature with Date	**Signature with Date**
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Prof. Neethu KN	**Dr. K M Sudarshan**
Guide	**Director, School of ECE**

	Name of the Examiner with affiliation	**Signature with Date**
1.	____________________	____________________
2.	____________________	____________________

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Abstract

Industrial motors are one of the most important components used in manufacturing industries, automation systems, and production plants. Sudden motor failures can lead to production downtime, financial losses, and safety risks. Traditional maintenance methods are inefficient because they either wait for failure to occur or follow fixed maintenance schedules without considering the actual condition of the motor.

This project presents a Predictive Maintenance System for Industrial Motors using Multi Sensor Data and Machine Learning. The proposed system continuously monitors motor health using multiple sensors such as LM35 temperature sensor, SW420 vibration sensor, and IR sensor for RPM measurement. Sensor data is collected using an ESP32 microcontroller and analyzed using a Machine Learning model to predict possible motor faults before complete failure occurs.

The system classifies motor conditions into NORMAL, OVERHEAT, UNBALANCE, and STOPPED states based on sensor values. Real-time data is transmitted to Firebase Realtime Database through Wi-Fi communication and visualized using a dashboard created in Firebase Studio.

The proposed system helps industries reduce downtime, improve operational efficiency, decrease maintenance costs, and increase equipment reliability. The project combines Embedded Systems, IoT, Cloud Computing, and Machine Learning technologies to create an intelligent industrial monitoring system. The Random Forest classifier achieved an accuracy of approximately 85%–95% during testing.

Chapter-1
Introduction

Industrial motors are widely used in industries for manufacturing, automation, robotics, conveyor systems, pumps, and compressors. Continuous operation of motors under varying loads and environmental conditions may lead to overheating, excessive vibration, imbalance, and mechanical wear. Unexpected motor failure can stop the entire industrial process and result in major economic losses.

Traditional maintenance approaches include reactive maintenance and preventive maintenance. Reactive maintenance repairs equipment only after failure occurs, while preventive maintenance follows fixed schedules without considering actual machine condition. Both approaches increase maintenance cost and reduce efficiency.

Predictive maintenance is an advanced maintenance strategy that continuously monitors machine health and predicts failures before breakdown occurs. This project implements an IoT-based predictive maintenance system using sensors and Machine Learning.

The proposed system uses:

- LM35 temperature sensor
- SW420 vibration sensor
- IR sensor for RPM measurement
- ESP32 microcontroller
- Firebase cloud database
- Machine Learning model (Random Forest Classifier)

The collected sensor data is analyzed using a Random Forest Machine Learning algorithm to classify motor health conditions. A real-time monitoring dashboard is developed using Firebase Studio for visualization and alerts.

1.1 Problem Statement

Industrial motors often fail unexpectedly due to various operational and environmental factors. Such unexpected failures disrupt the entire production process and impose heavy financial and safety burdens on industries.

Common causes of motor failure include:

- Overheating
- Excessive vibration
- Shaft imbalance
- Mechanical wear
- Load fluctuations

Such failures result in:

- Production downtime
- Increased maintenance expenses
- Reduced industrial productivity
- Safety hazards to personnel and equipment

Traditional maintenance systems cannot provide accurate real-time motor condition monitoring. Therefore, industries require an intelligent system that continuously monitors motor parameters, detects abnormal conditions early, predicts possible faults, sends real-time updates and alerts, and reduces downtime and maintenance costs.

1.2 Objective

The major objectives of this project are:

- To study predictive maintenance techniques for industrial motors
- To monitor motor health using multi sensor data
- To measure temperature, vibration, and RPM continuously
- To analyze motor condition using Machine Learning
- To predict faults before complete motor failure
- To upload real-time data to Firebase cloud
- To develop a monitoring dashboard using Firebase Studio
- To reduce industrial downtime and maintenance costs
- To achieve approximately 85%–95% classification accuracy using Random Forest algorithm

Chapter-2
Literature survey

A detailed study of IEEE research papers was carried out to understand different predictive maintenance techniques used in industries. The following table summarizes the key papers reviewed:

Paper	Method Used	Main Contribution	Limitation
Predictive Maintenance using IoT	IoT + ML	Real-time monitoring	Complex cloud architecture
LSTM-based Predictive Maintenance	LSTM Neural Network	Early fault prediction	Requires large datasets
ANN-based Motor Fault Detection	Artificial Neural Network	High accuracy fault detection	Complex signal processing
Motor Current Signature Analysis	MCSA	Electrical fault detection	Requires DSP techniques
Predictive Maintenance using Load Analysis	Load Monitoring	Detects shaft imbalance	Limited parameter analysis
Condition Based Monitoring Review	CBM Techniques	Overview of CBM methods	No practical implementation
Random Forest based Monitoring	Random Forest	Simple fault classification	Limited real-time integration

Research Gap

Most existing systems:

- Require expensive industrial hardware
- Use complicated signal processing techniques
- Need large datasets for training
- Lack simple real-time monitoring dashboards

The proposed system focuses on:

- Low-cost implementation using readily available sensors
- Real-time IoT monitoring through Firebase cloud
- Easy deployment using ESP32 microcontroller
- Multi-sensor monitoring for comprehensive fault detection
- Machine Learning based prediction using Random Forest

Chapter-3
Proposed Work

3.1 Methodology

The proposed system is an IoT and Machine Learning based predictive maintenance system for industrial motors. It integrates sensor hardware, embedded computing, cloud connectivity, and artificial intelligence to deliver a comprehensive motor health monitoring solution.

Features of Proposed System:

- Real-time motor monitoring using multiple sensors
- Multi-sensor data collection (temperature, vibration, RPM)
- Machine Learning based fault prediction using Random Forest
- Cloud-based monitoring through Firebase Realtime Database
- Real-time dashboard visualization using Firebase Studio
- Early warning alerts for abnormal conditions

Parameters Monitored:

- Temperature (via LM35 sensor)
- Vibration (via SW420 sensor)
- RPM – Revolutions per Minute (via IR sensor)

The system uses ESP32 to collect sensor data and transmit it to the Firebase cloud database through WiFi. A Random Forest model embedded in the ESP32 predicts motor conditions locally, and results are also synchronized to the cloud.

3.2 Block Diagram

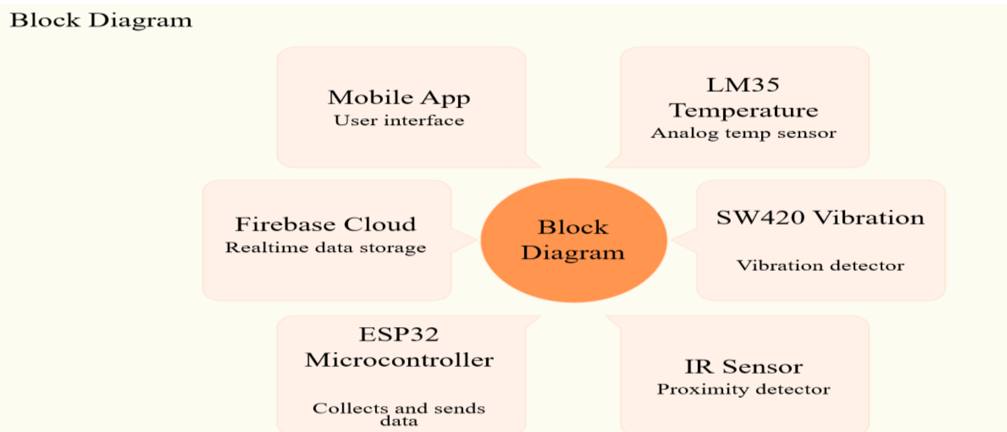


Fig 3.2.1 Block Diagram of the System

3.3 Flow Chart

The following flowchart represents the complete operational flow of the predictive maintenance system:

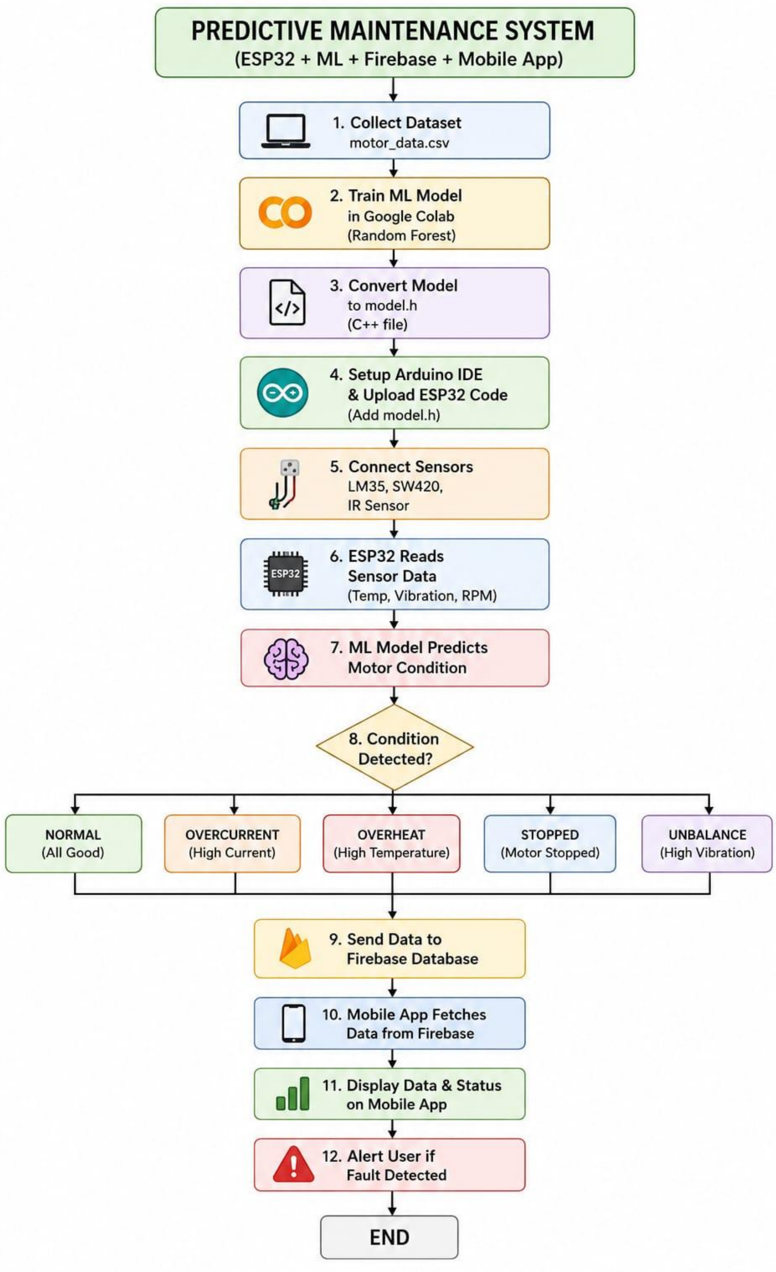


Fig 3.3.1 Flowchart of Data flow

3.4 Hardware and Software Descriptions

3.4.1 Hardware Requirements

Component	Quantity	Purpose
ESP32 Microcontroller	1	Main controller with WiFi capability
LM35 Temperature Sensor	1	Measures motor surface temperature
SW420 Vibration Sensor	1	Detects vibration and mechanical shocks
IR Sensor	1	Measures motor shaft RPM
Breadboard	1	Circuit assembly and prototyping
Jumper Wires	Multiple	Electrical connections between components
USB Cable	1	Power supply and programming interface

3.4.2 Software Requirements

Software / Tool	Purpose
Arduino IDE	ESP32 firmware programming and code uploading
Google Colab	Machine Learning model training and testing
Firebase Realtime Database	Cloud storage and real-time data synchronization
Firebase Studio	Dashboard development and visualization
Python	Data preprocessing and model development

3.4.3 Python Libraries Used:

- pandas – Data manipulation and preprocessing
- numpy – Numerical computation
- scikit-learn – Machine Learning model implementation
- micromlgen – Conversion of ML model to C++ for ESP32 deployment

3.5 Sensors Descriptions

3.5.1 LM35 Temperature Sensor

The LM35 is a precision analog temperature sensor used to monitor motor surface temperature. It produces a linear output voltage proportional to temperature, making it easy to interface with the ESP32's ADC (Analog to Digital Converter).

Key Features:

- High accuracy temperature measurement
- Linear output voltage of 10 mV/°C
- Low power consumption
- Operating range: -55°C to +150°C

3.5.2 SW420 Vibration Sensor

The SW420 is a vibration detection module that detects mechanical vibrations and shocks. When the motor experiences excessive vibration due to imbalance or bearing faults, the sensor triggers a digital output signal that is read by the ESP32.

Applications:

- Mechanical fault detection
- Shaft imbalance detection
- Bearing wear and damage monitoring

3.5.3 IR Sensor (RPM Measurement)

The IR sensor is used for non-contact RPM measurement of the motor shaft. A reflective tape is attached to the rotating shaft. As the shaft rotates, each pass of the tape generates a reflected infrared pulse detected by the sensor.

RPM is calculated using the formula:

$$\mathbf{RPM = Pulses\ Per\ Second \times 60}$$

3.6 System Architecture

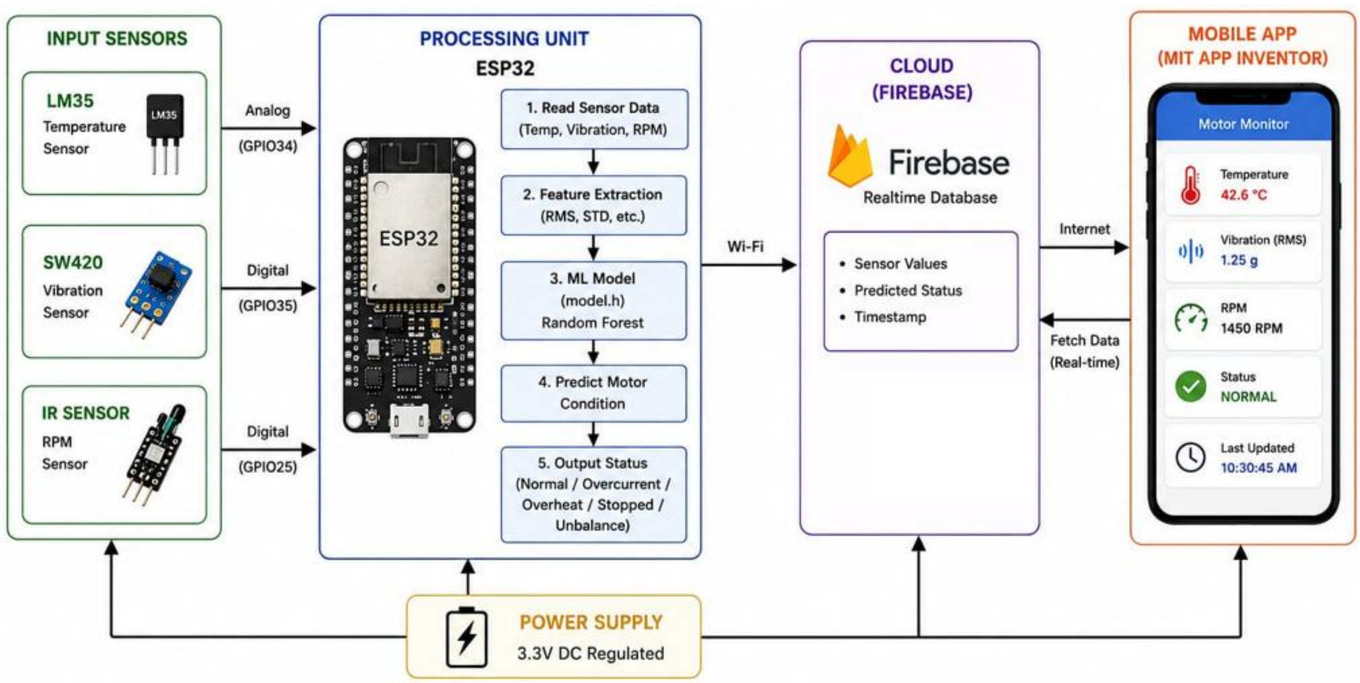
The system architecture consists of five interconnected layers, each responsible for a specific function in the data processing and monitoring pipeline.

Architecture Layers:

- Sensor Layer – LM35, SW420, and IR sensors collect raw physical data from the motor
- Processing Layer – ESP32 microcontroller digitizes and processes sensor readings
- Machine Learning Layer – Random Forest model predicts motor condition from processed data
- Cloud Layer – Firebase Realtime Database stores and synchronizes data via WiFi
- Dashboard Layer – Firebase Studio displays live sensor values and motor status

Data Flow:

Sensors → ESP32 → ML Prediction → Firebase → Firebase Studio Dashboard



3.7 Working Principle

The system operates in a continuous loop, collecting sensor data, analyzing it using the embedded Machine Learning model, and uploading results to the cloud for real-time monitoring. The step-by-step working is as follows:

1. Sensors continuously collect motor parameters including temperature, vibration, and rotation.
2. ESP32 reads and digitizes the analog and digital sensor values.
3. IR sensor calculates motor shaft RPM using pulse counting.
4. Sensor values are processed by the embedded Machine Learning model.
5. Random Forest algorithm predicts the motor condition class.
6. Sensor data and prediction results are uploaded to Firebase Realtime Database.
7. Firebase Studio dashboard displays live values and motor health status.
8. Alerts are generated automatically during abnormal motor conditions.

3.8 Machine Learning Model

3.8.1 Overview

Machine Learning enables the system to automatically learn patterns from sensor data and predict motor conditions without explicit programming. The system was trained on labeled sensor data representing different motor states.

3.8.2 Algorithm Used: Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions for improved accuracy and robustness. It was chosen for this project due to its simplicity, high accuracy, and ability to be converted to C++ for ESP32 deployment.

Input Features:

- vib_rms – Root Mean Square of vibration signal
- vib_std – Standard deviation of vibration signal
- rpm – Motor shaft revolutions per minute
- temp_C – Motor temperature in degrees Celsius

Output Classes:

- NORMAL – Motor operating within acceptable parameters
- OVERHEAT – Motor temperature exceeds safe threshold
- UNBALANCE – Excessive vibration indicating shaft imbalance
- STOPPED – Motor not rotating or rotating below minimum RPM

3.8.3 Training Process

1. Data Collection – Sensor readings collected under different motor conditions
2. Data Preprocessing – Noise filtering, normalization, and labeling
3. Feature Extraction – Computing statistical features from raw sensor data
4. Model Training – Random Forest trained using Google Colab and scikit-learn
5. Model Deployment – Model converted to C++ using micromlgen and deployed on ESP32

Model Accuracy: Approximately 85%–95% accuracy was achieved using the Random Forest algorithm during testing on the validation dataset.

3.9 Firebase Integration

Firebase Realtime Database is used as the cloud backend for storing and synchronizing sensor data in real time. The ESP32 communicates with Firebase over WiFi using the Firebase REST API, uploading sensor readings and prediction results after each measurement cycle.

Advantages of Firebase:

- Real-time data updates with minimal latency
- Secure remote monitoring from anywhere
- Free-tier availability suitable for prototype deployment
- Fast read/write performance

Parameters Stored in Firebase:

- Temperature (°C)
- Vibration (RMS and standard deviation)
- RPM value
- Motor health status (NORMAL / OVERHEAT / UNBALANCE / STOPPED)
- Timestamp of each measurement

3.10 Dashboard Monitoring Using Firebase Studio

The monitoring dashboard was developed using Firebase Studio, which provides a visual interface for displaying real-time data from the Firebase Realtime Database. The dashboard enables remote monitoring of the motor from any internet-connected device.

Dashboard Features:

- Real-time display of temperature, vibration, and RPM values
- Motor status indicator (NORMAL / OVERHEAT / UNBALANCE / STOPPED)
- Live data updates directly from Firebase cloud
- Fault indication for abnormal conditions
- Remote monitoring accessible from mobile and desktop devices

Firebase Studio simplifies cloud dashboard development and significantly improves accessibility for remote industrial monitoring. The dashboard auto-refreshes as new data arrives from the ESP32.

3.11 Algorithms

Algorithm Steps:

1. Start system
2. Initialize sensors and ESP32 peripherals
3. Connect ESP32 to WiFi network
4. Read sensor values: temperature, vibration, RPM pulses
5. Calculate RPM from pulse count
6. Extract features: vib_rms, vib_std, rpm, temp_C
7. Process data using embedded ML model
8. Predict motor condition: NORMAL / OVERHEAT / UNBALANCE / STOPPED
9. Upload sensor values and prediction to Firebase
10. Display values and status in Firebase Studio dashboard
11. Generate alert if abnormal condition detected

    Repeat continuously from Step 4

Chapter-4
Result Analysis

The proposed predictive maintenance system was implemented and tested on an industrial motor prototype. The system successfully monitored motor conditions in real time with consistent accuracy.

4.1 Achievements:

- Successful real-time sensor monitoring with continuous data acquisition
- Real-time cloud updates to Firebase with minimal latency
- Accurate ML prediction using Random Forest classifier
- Mobile app and web dashboard visualization via Firebase Studio
- Fault alert generation for abnormal motor conditions

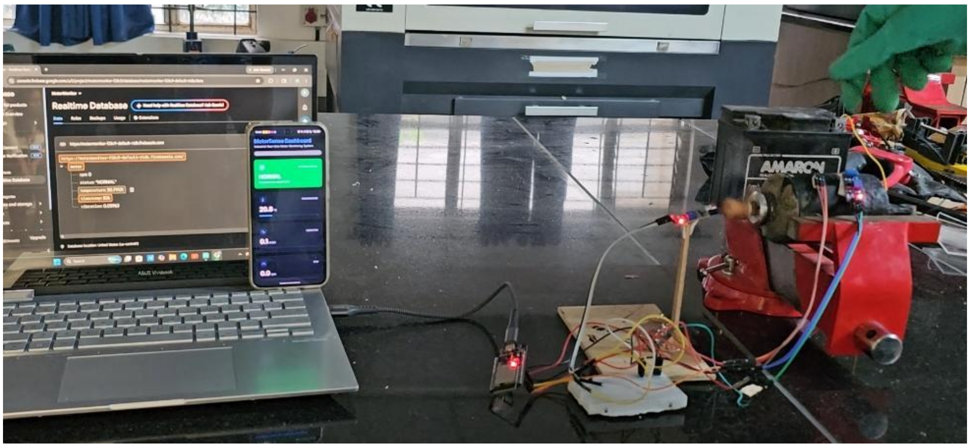
Model Accuracy:

Approximately 85%–95% classification accuracy was achieved using the Random Forest algorithm during validation testing.

Observed Results:

- Increase in vibration RMS above threshold → UNBALANCE fault detected
- Temperature rise above safe limit → OVERHEAT condition flagged
- RPM falling to zero or near zero → STOPPED condition identified
- All parameters within range → NORMAL classification confirmed

The system demonstrated efficient and reliable motor condition monitoring using multi-sensor data, and the dashboard provided intuitive, real-time access to motor health information from remote locations.



4.1.1 Result screenshot

4.2 Advantages

- Early fault detection prevents catastrophic motor failures
- Reduced production downtime through predictive intervention
- Low maintenance cost compared to traditional approaches
- Real-time monitoring enables immediate response to faults
- Cloud-based access allows remote monitoring from anywhere
- Remote monitoring capability reduces need for on-site inspection
- Easy deployment using low-cost, commercially available components
- Low-cost implementation suitable for small and medium industries
- Scalable architecture allows future expansion to multiple motors

4.3 Application

- Industrial motor monitoring in manufacturing plants
- Smart factories adopting Industry 4.0 concepts
- Automation industries using conveyor systems and robotic arms
- Manufacturing plants with high-value production equipment
- Conveyor and material handling systems
- Industrial IoT (IIoT) systems and deployments
- Predictive maintenance platforms for multiple equipment types
- Pump and compressor monitoring in process industries

Chapter 5
Conclusion & Future work

5.1 Conclusion

The Predictive Maintenance of Industrial Motors Using Multi Sensor Data project successfully demonstrated real-time motor monitoring using IoT and Machine Learning technologies. The system monitors critical motor parameters temperature, vibration, and RPM using sensors connected to the ESP32 microcontroller.
Sensor data is analyzed by an embedded Random Forest Machine Learning model that classifies motor health into four conditions: NORMAL, OVERHEAT, UNBALANCE, and STOPPED. Firebase cloud integration and Firebase Studio dashboard enable remote monitoring and real-time visualization of motor health status.

The system achieved approximately 85%–95% classification accuracy and successfully generated fault alerts for abnormal motor conditions. The project provides an efficient, low-cost, and intelligent solution for industrial predictive maintenance.

This project demonstrates the practical implementation of Industry 4.0 concepts, combining Embedded Systems, IoT, Cloud Computing, and Machine Learning to improve industrial reliability, reduce downtime, and prevent sudden motor failures.

5.2 Future Scope

The current system provides a strong foundation for further development. Future enhancements may include:

- Addition of current and voltage sensors for electrical fault detection
- Implementation of advanced Deep Learning algorithms such as LSTM for improved accuracy
- Simultaneous monitoring of multiple motors from a single dashboard
- GSM/SMS alert system for offline fault notifications
- Advanced web analytics dashboard with historical trend visualization
- Industrial-scale deployment across factory floors
- Integration with Industry 4.0 systems and digital twin platforms
- Edge computing optimization for faster on-device prediction

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